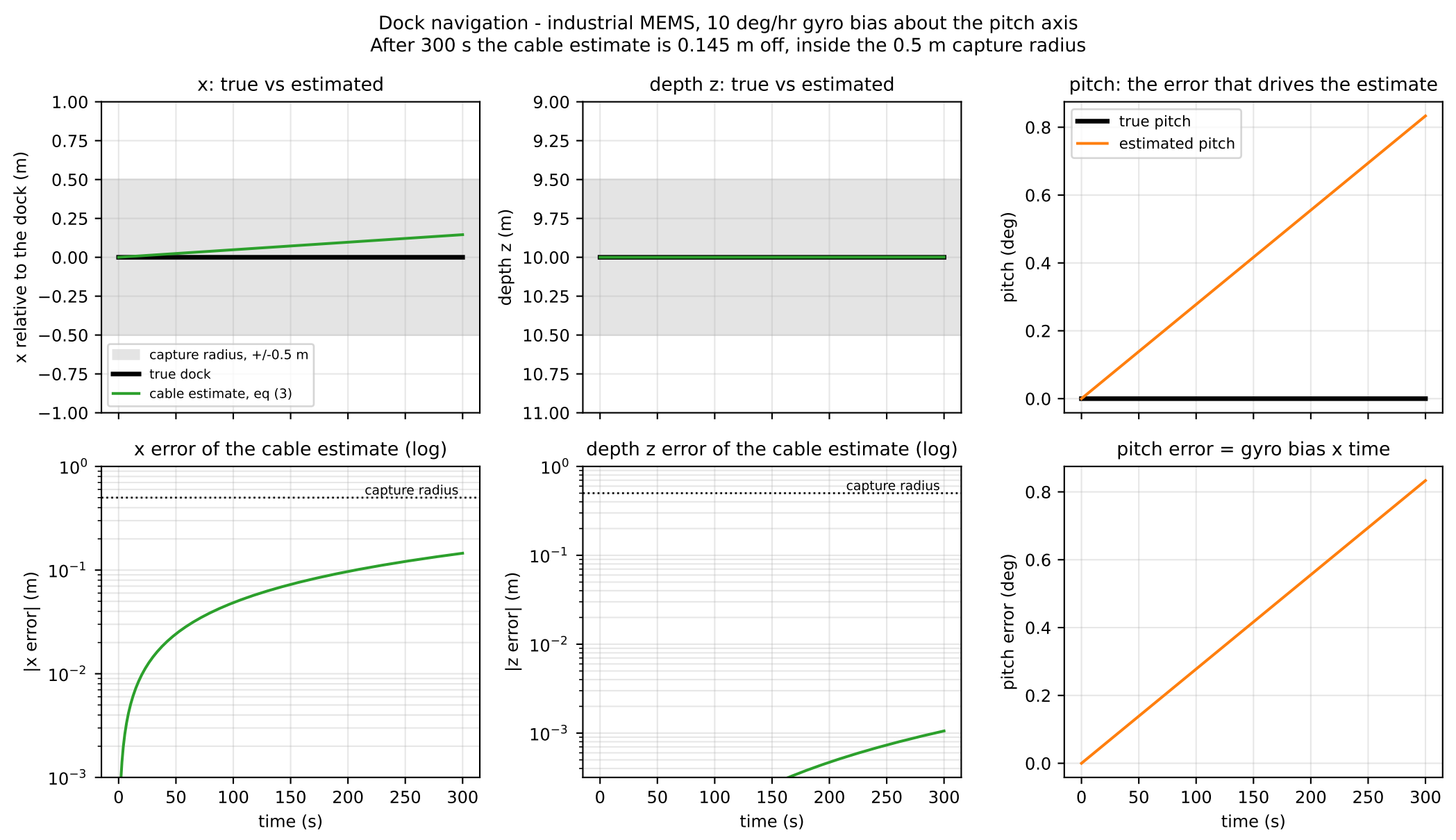
Dock navigation - industrial MEMS, 10 deg/hr gyro bias about the pitch axis
After 300 s the cable estimate is 0.145 m off, inside the 0.5 m capture radius
x: true vs estimated
x relative to the dock (m)
capture radius, +/-0.5 m
true dock
cable estimate, eq (3)
depth z: true vs estimated
depth z (m)
pitch: the error that drives the estimate
pitch (deg)
true pitch
estimated pitch
x error of the cable estimate (log)
|x error| (m)
capture radius
time (s)
depth z error of the cable estimate (log)
|z error| (m)
capture radius
time (s)
pitch error = gyro bias x time
pitch error (deg)
time (s)
Only x, depth and pitch are shown: a pitch-axis bias leaves y, roll and yaw exact. The vehicle and buoy have ideal IMUs in this scenario, so their estimates equal the truth.